Part 1. Using Public Key Algorithms

1) Role 1: You are the Sender

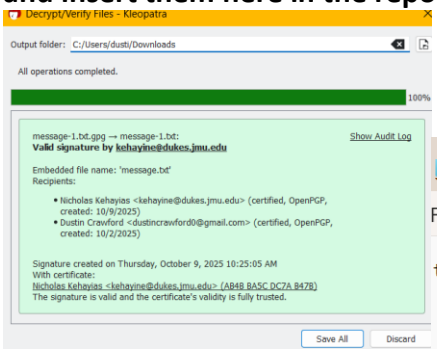
Encrypt a message for the receiver assigned to you (you need to choose your teammate). Note that you are the sender. In the “Public Key Repository Discussion” reply to the receiver’s post and attach the encrypted file in your reply. Please make sure that your message is appropriate. **Insert your message to the receiver in plaintext here (in the report).**

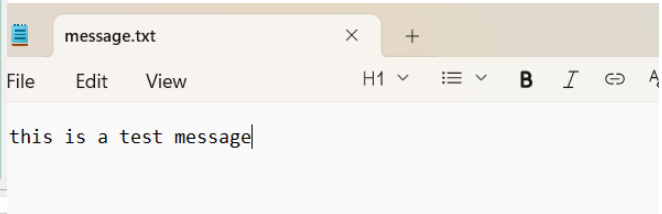
My message said ‘this is a test message’

2) Role 2: You are the Receiver

Decrypt the message that was sent you (attached by the sender in the reply to your post in “Public Key Repository Discussion”).

Take screenshots of the verification window and the decrypted message that was sent to you and insert them here in the report.

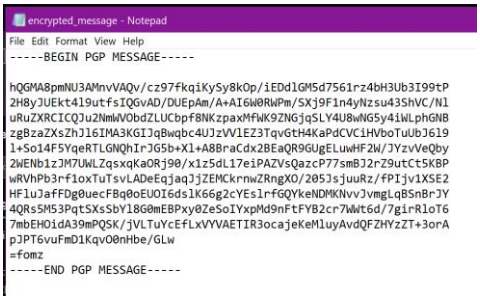


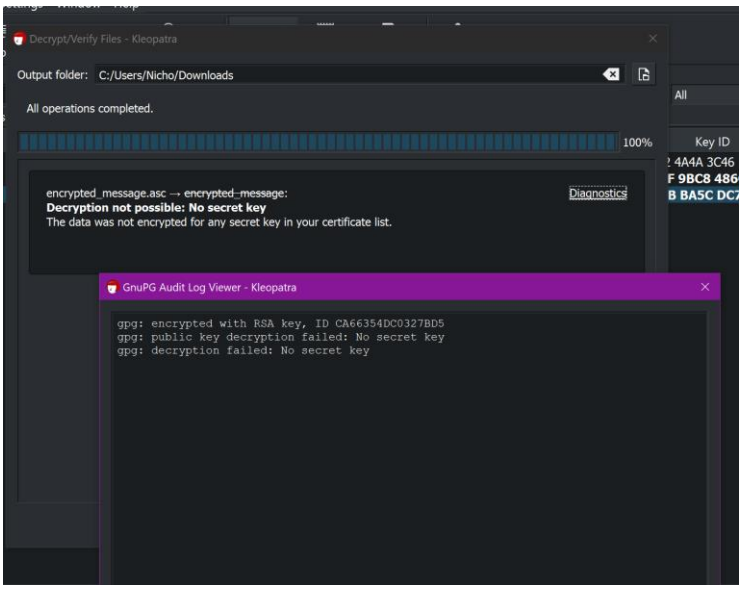


3) Role 3: You are the attacker

Pick an encrypted message that was not intended for you. Try to figure out what the message is. Explain your experience attempting to break the code.

When I first try to break the code just using Kleopatra, I am unable to read it due to lack of a private key. I also tried to open the message in a text editor to see if anything was recognizable, but it was not. Confirming online, I am unable to view the message unless I have the private key.





Part 2. Verifying Integrity of a Download

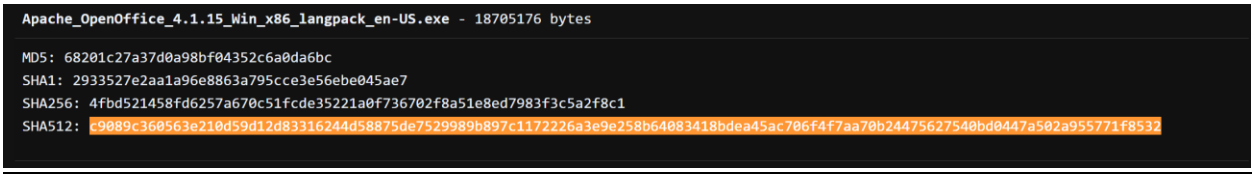
In this assignment you will verify that a file you download has not been changed during download.

Go to the site https://www.openoffice.org/download/index.html.

Note that there two download options: “Download full installation” and “Download language pack.” Download the language pack only and do NOT install it.

Note that two hash values are given for the language pack on the site: SHA 256 and SHA512.
Copy and paste these values in this document.
256 - 4fbd521458fd6257a670c51fcde35221a0f736702f8a51e8ed7983f3c5a2f8c1
512 - c9089c360563e210d59d12d83316244d58875de7529989b897c1172226a3e9e258b64083418bdea45ac706f4f7aa70b24475627540bd0447a502a955771f8532

Use an online hash calculator to calculate the SHA256 and SHA512 hashes for the downloaded file. You can use https://md5file.com/calculator.
Insert screenshots of these values in this document as well.



What to submit

Insert screenshots from above steps.
In the Word file, include your answers to the following questions with a different color:

1. Compare these hash values on the site with the hash values you have calculated. Are they the same? Explain.
   They are the same. This indicates that the file’s integrity has been preserved. The calculations from the author match those of the file I downloaded indicating nothing has been modified.
2. What is your conclusion if the SHA256 hash value provided on the site is the same as the one you calculated?
   The downloaded file is identical to the one the author intended to distribute. No bits have been altered.

3. What is your conclusion if the SHA256 hash value provided on the site is different from the one you calculated?
   It was not different from the one I calculated. However, if the values did differ, the file was likely tampered with, corrupted, or incorrectly downloaded.
4. If the provided hash value is the same as the one you calculated, does that mean that this is absolutely the right file that delivers what it promises, or could it be malicious? Explain.
   No, it could still be malicious. If the original file itself was malicious, the hash will still match.
5. How could a second hash value (such as SHA512, in this case) help to verify the integrity of the download? Explain.
   Using the second has value makes tampering even more difficult because an attacker would need to tamper with the file in a way that matches both hash algorithms simultaneously.

Grading Rubric

Grading Rubric:

Criteria		Points
Part 1	Public key in “Public Key Repository” Discussion	1
	Complete the task for Role 1	3
	Complete the task for Role 2	3
	Complete the task for Role 3	3
Part 2	Screenshots of four hash values	2
	Answers to questions about hash functions	7
	Grammer/Spells	1
Total		20